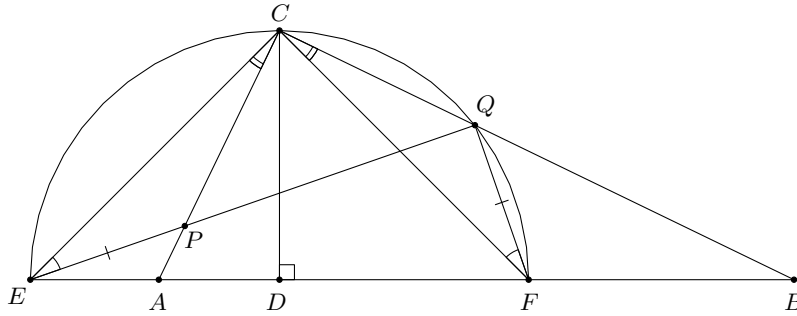


Problem 81. Let ABC be a triangle with a right angle at C and $AC < BC$. Let D be the foot of the perpendicular from C to AB . The circle centred at D with radius CD intersects BC at Q , and intersects the line AB at two distinct points E and F with F lying between A and B . Let QE intersect AC at P . Prove that $PE = QF$.

Solution.



Since, by definition, EF is the diameter of the circle centred at D with radius CD , we see that $\triangle EFC$ is isosceles and so $EC = FC$. Moreover, in this circle, we have $\angle CEP = \angle CEQ = \angle CFQ$. Now, since $\angle ECF = 90^\circ = \angle ACB$, we have

$$\angle ECP = \angle ECA = 90^\circ - \angle ACF = \angle FCB = \angle FCQ,$$

and so $\triangle EPC \cong \triangle FQC$. Therefore $PE = QF$. □

Problem 82. Find all pairs of positive integers (m, n) satisfying the following equation:

$$(m+1)^{m-1} + (m-1)^{m+1} = n^n.$$

Solution. The answers are $(m, n) = (1, 1), (2, 2)$.

First observe that we have

- $n^n = (m+1)^{m-1} + (m-1)^{m+1} < 2(m+1)^{m+1} < (m+2)^{m+2}$; and
- $n^n = (m+1)^{m-1} + (m-1)^{m+1} \geq (m+1)^{m-1} \geq (m-1)^{m-1}$.

Now we consider the following cases.

Case 1: If $n = m - 1$, then $n^n = (m+1)^{m-1} + (m-1)^{m+1} > 0 + (m-1)^{m+1} \geq (m-1)^{m-1}$, which is impossible.

Case 2: If $n = m$, then

$$m^m = (m+1)^{m-1} + (m-1)^{m+1}. \quad (*)$$

- (1) When $m = 1$, $(*)$ holds and so $(m, n) = (1, 1)$ is a solution.
- (2) When $m = 2$, $(*)$ holds and so $(m, n) = (2, 2)$ is another solution.
- (3) When $m = 3$, $(*)$ does not hold and so there are no solutions in this case.
- (4) When $m \geq 4$, we note that $f(x) = \frac{\ln x}{x}$ is strictly decreasing for $x > e$ because $f'(x) = \frac{1 - \ln x}{x^2} < 0$ for $x > e$, and so $(m-1)^m > m^{m-1}$. Hence

$$\begin{aligned} (m+1)^{m-1} + (m-1)^{m+1} &= (m+1)^{m-1} + (m-1)(m-1)^m \\ &> (m+1)^{m-1} + (m-1)m^{m-1} \\ &> m^{m-1} + (m-1)m^{m-1} \\ &= m^m, \end{aligned}$$

and so there are no solutions in this case as well.

Case 3: If $n = m + 1$, then $(m+1)^{m+1} = (m+1)^{m-1} + (m-1)^{m+1}$. Hence

$$(m-1)^{m+1} = (m+1)^{m-1} ((m+1)^2 - 1) = (m+1)^{m-1} m(m+2),$$

which clearly is impossible because $\gcd(m-1, m) = 1$ and $m+2, m+1 > m-1$ and so there are no solutions in this case too.

Therefore the only pairs of positive integers are $(m, n) = (1, 1), (2, 2)$. □

Problem 83. A *knight* is a piece in the game of chess. It is known that a knight is allowed to move to a square on the chessboard that is two squares away horizontally and one square vertically, or two squares vertically and one square horizontally (i.e. the complete move therefore looks like the letter ‘L’).

An *open knight’s tour* is a sequence of moves of a knight on a chessboard such that the knight visits every square exactly once.

Find all positive integers $n \geq 3$ for which there exists an open knight’s tour on a $3 \times n$ chessboard.

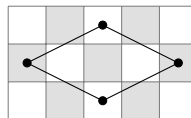
(Note that the knight may start at any square on the chessboard and may not necessarily end at the square where it started.)

Solution. An open knight’s tour exists on a $3 \times n$ chessboard for $n = 4$ or $n \geq 7$.

Below we will simply use the terms board and tour to mean the terms chessboard and open knight’s tour, respectively.

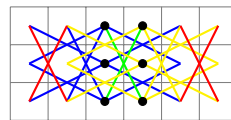
First we show that no tours exist when $n = 3, 5, 6$.

- For a 3×3 board, there is no tour because, from the square situated in the center, there is no way the knight can move.
- For a 3×5 board, we note that there is an odd number of squares. So if we use the usual colouring of the chessboard into two colours of white and black/gray, and assume that the upper left is white, we see that a tour in this board must have its ends in the white squares because there are more white squares!



If the knight moves to the square on the second row and first column, or moves to square on the second row and fifth column, which are both black/gray, then observe that either of these squares will never be ends of the tour. Consequently this forces the knight to move in such a way as to form a cycle of 4 moves! Hence the desired tour for a 3×5 board cannot exist.

- For a 3×6 board, observe that, by counting the knight’s possible moves, there are 22 moves incident with the two middle columns (and for ease of counting, the moves are coloured, 10 blue edges, 10 yellow edges and 2 green edges, as shown in the board below) and only 4 other moves (4 red edges). Now if a tour exists, it would use $3 \times 6 - 1 = 17$ moves, at most two on each square of the middle columns, giving a total of at most 12 moves, leaving at least 5 more required moves. But only 4 moves are available. Hence no tours exist for a 3×6 board.



Next we will prove that a tour exists on the remaining $3 \times n$ boards by induction. For the base step, we show a tour beginning in the upper left for each of the boards with $n = 4, 7, 9, 10$. The positions of the knight’s moves are labelled numerically.

1	4	7	10
8	11	2	5
3	6	9	12

3×4 board

1	14	17	20	11	8	5
16	19	12	3	6	21	10
13	2	15	18	9	4	7

3×7 board

1	14	17	10	7	4	19	22	25
16	9	12	3	18	23	26	5	20
13	2	15	8	11	6	21	24	27

3×9 board

1	4	7	22	15	20	13	26	29	18
8	23	2	5	10	25	16	19	12	27
3	6	9	24	21	14	11	28	17	30

3×10 board

Now for the inductive step, we add two boards, each with a tour beginning in the upper left,

(1) by simply putting them side by side; and

(2) by suitably renumbering the part of the second board in the outcome, i.e. continuing with the numbering of the first board.

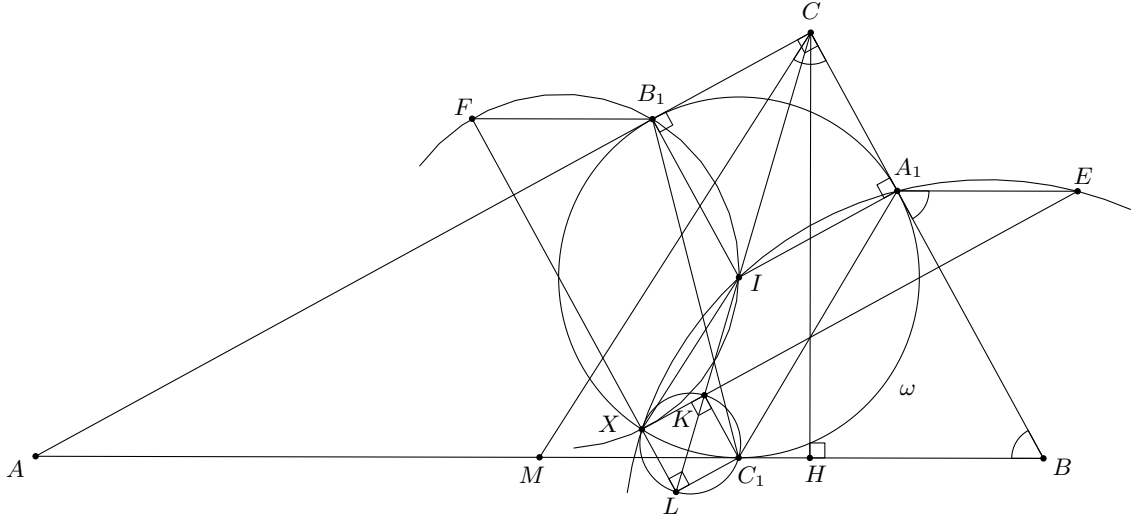
It is clear that adding in this way produces a tour because the knight at the end position of each board can always move to the upper left corner of the next board. Below is an example of adding a 3×4 board to a 3×7 board:

1	4	7	10		1	14	17	20	11	8	5		1	4	7	10	13	26	29	32	23	20	17
8	11	2	5		16	19	12	3	6	21	10		8	11	2	5	28	31	24	15	18	33	22
3	6	9	12		13	2	15	18	9	4	7		3	6	9	12	25	14	27	30	21	16	19

To finish we add the 3×4 board with a tour beginning in the upper left and ending in the lower right to the left of a $3 \times n$ board with a tour beginning in the upper left to obtain a $3 \times (n + 4)$ board with a tour beginning in the upper left. Since $n = 4, 7, 9, 10 \equiv 0, 3, 1, 2 \pmod{4}$, respectively, the result follows by induction. $\square$

Problem 84. Let ABC be a triangle with a right angle at C , let I be its incentre, and let H be the foot of the perpendicular from C to AB . The incircle ω of the $\triangle ABC$ is tangent to the sides BC , CA , and AB at A_1 , B_1 , and C_1 , respectively. Let E and F be the reflections of C in the lines C_1A_1 and C_1B_1 , respectively, and let K and L be the reflections of H in the same lines. Prove that the circles A_1EI , B_1FI , and C_1KL have a common point.

Solution.



The line C_1A_1 is parallel to the external angle bisector of $\angle B$, so the reflection in C_1A_1 maps the segment A_1C to the segment A_1E parallel to AB . Similarly, $B_1F \parallel AB$. Notice also that $A_1E = A_1C = B_1C = B_1F = r$, where r is the inradius of $\triangle ABC$.

Let M be the midpoint of AB . Let X be the point of ω such that $IX \parallel CM$ and X is closer to M than to C . Notice that

$$\angle EA_1I = 90^\circ + \angle EA_1B = 90^\circ + \angle CBM = 90^\circ + \angle BCM = \angle A_1IX,$$

and $A_1E = IA_1 = IX$; thus, XIA_1E is an isosceles trapezoid. Hence X lies on the circle IA_1E , and $EX \parallel A_1I$. Similarly, X lies on the circle IB_1F , and $FX \parallel B_1I$. It remains to show that X lies on the circle C_1KL .

Under the symmetry in C_1A_1 , the line CH (perpendicular to AB) maps to the line through E perpendicular to BC , i.e. CH maps to EX . Therefore, the projection H of C_1 onto CH maps to the projection K of C_1 onto EX . Similarly, L is the projection of C_1 onto FX . So the quadrilateral C_1KXL is cyclic, due to right angles at K and L . $\square$